Pramesh Sharma

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Skills

- Python
- JavaScript
- C++
- Go
- TensorFlow
- PyTorch
- Google Cloud
- Docker
- Linux
- Git

Awards

Regents Scholar

AvenueE Scholar

Engineering, Math, and Physics Scholar

Michael Sager Memorial Scholar

Deloitte Leadership, Allyship & Mentorship Program

Genentech gLINX Program

Education

University of California, Davis — Davis, CA
Bachelor of Science (Computer Science) — 09/2021 - 06/2024

Experience

National Institutes of Health — Bethesda, MD
Postbaccalaureate Fellow — 08/2025 - Present

- Developed an AI interpretability tool leveraging AlphaFold attention weights to extract biologically meaningful insights from protein structures, earning 10+ GitHub stars and a bioaRxiv manuscript
- Engineered an end-to-end inference pipeline integrating distance-based clustering and backbone geometry to identify fold-switching proteins
- Streamlined automated testing and multi-environment package releases with GitHub Actions, reducing deployment time by 3 minutes per pull request
- Instituted an elevated standard for code quality increasing test coverage to 90% using pytest on a codebase spanning over 50+ integration and unit tests

Intelligent Retail Lab by Walmart — Sunnyvale, CA
Software Engineer — 07/2024 - 07/2025

- Optimized collection of hardware metrics across cluster of NVIDIA Jetson GPUs by replacing jetson-stats through development of a real-time system monitoring tool using Python, Prometheus, and Grafana
- Automated machine learning training workflows by developing a model converter that transforms PyTorch ONNX models into DeepStream-compatible TensorRT engines using Kubeflow and WandB
- Improved image transformation processing for 40,000+ grocery item images by building a scalable data augmentation pipeline in Python using Apache Beam and Dataflow and deployed as a Google Flex Template
- Reduced object detection inference latency across edge cameras by deploying a scalable DeepStream pipeline to run multiple YOLO models via gRPC streaming of inference outputs to Google BigQuery for cross-run tracking

Publications

EasyDIVER+: an advanced tool for analyzing high throughput sequencing data from in vitro evolution of nucleic acids or amino acids
Journal of Molecular Evolution
Celia Blanco, Allison Tee, Pramesh Sharma, Matilda S. Newton, Kun-Hwa Lee, Samuel E. Erickson, Burckhard Seelig, Irene A. Chen

AMMPER-2: A spatially explicit agent-based model of microbial radiobiology with redox dye simulation
American Society for Microbiology
Jessica A. Lee, Daniel Palacios, Amrita Singh, Madeline Marous, Pramesh Sharma, Lauren Liddel, Diana Gentry, Sergio Santa Maria

Internships

NASA

Computational Biology Intern

- Conducted radiobiology literature review to inform integration of the Linear Quadratic model into AMMPER, aligning radiation dose-response with experimental data and improving biological relevance
- Redefined cell health model using Python to simulate and enhance cell behavior predictions, establishing probability distributions for apoptosis, DNA damage, and mitotic catastrophe
- Enhanced model accuracy by 40% by designing an AlamarBlue viability model using Michaelis-Menten kinetics, leveraging AutoML and SMAC3; optimized hyperparameters using Grid and Bayesian search strategies

NASA

Space Systems Engineering Intern

- Streamlined cross-functional collaboration by standardizing project modeling for the $14.8B Lunar Pilot Excavator through a Systems Engineering Modeling Management Plan in MagicDraw
- Replaced paper-based workflows across 3+ engineering divisions through a strategy aligning digital engineering review processes
- Communicated project modeling strategy to key stakeholders by presenting the SEMMP template and Concept of Operations to design and chief engineering teams

Genentech

Computational Proteomics Intern

- Increased efficiency of antibody analysis for wet lab scientists by developing an internal tool using the Pyramid REST framework that converts protein-peptide data into heat maps
- Streamlined data processing for large datasets by creating a command line interface using the optparse R package that allows users to handle up to 3,000 samples of protein-peptide information at once
- Deployed tool into production by containerizing dataMAPPs R package with Docker and Python, ensuring robust and scalable performance

Genentech

Pharma Technical Operations Intern

- Increased efficiency of drug campaign creation by optimizing manufacturing on-call service using JavaScript and PagerDuty API
- Automated user access to drug campaign incidents with JavaScript and Google Sheets API, resulting in cost-savings of 5-minutes per incident
- Standardized Genentech's PagerDuty response procedures by documenting a comprehensive 10-page Rapid Response Playbook ensuring SOP compliance